

Box Model:

The CSS Box Model is a foundational concept that defines the layout and structure of every HTML element displayed on a webpage. It's essentially a box wrapping around an element, determining how its dimensions, padding, border, and margin are calculated to control its placement and appearance.

Components of the CSS Box Model

Every HTML element is made up of four main layers, arranged from the innermost to the outermost:

1. Content

This is the core area of the box where the actual element data—such as text, images, or videos—is displayed.

The dimensions of this area are controlled using the width and height properties.

2. Padding

Padding is the clear space that surrounds the content area from the inside.

It provides internal spacing between the content and the element's border.

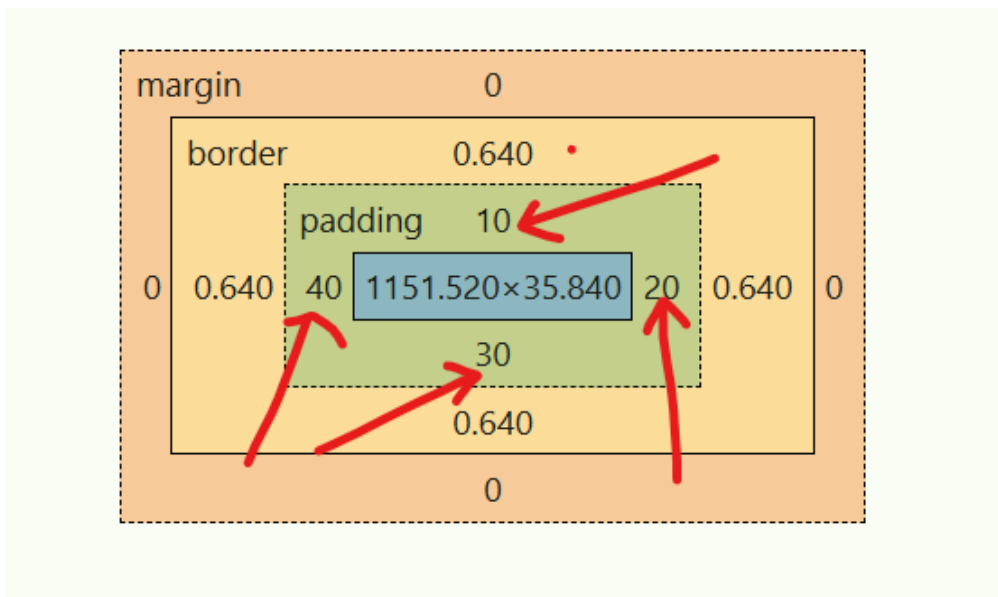
Padding increases the overall size of the element without affecting the content itself.

```
<style>
  div{
    padding-top: 10px;
    padding-right: 20px;
    padding-bottom: 30px;
    padding-left: 40px;
    border : 1px solid black;
  }
</style>
```

Padding shorthand :

```
<style>
  div{
    /* top right bottom left */
    padding : 10px 20px 30px 40px;
    border : 1px solid black;
  }
</style>
```

Verification of output :



3. Border

The border wraps around the padding and defines the outer edge of the element's box.

It can be styled using properties like border-width, border-style, and border-color (for example, 2px solid red).

Some of the Border Properties are :

property	value	description
border-width	Keyword -> Thin, medium, thick Absolute unit -> 4px, Relative unit-> 2em	sets the width of an element's border.
border-style	Dotted, dashed, solid, double, none, hidden,	specifies what kind of border to display
border-color	Name -> Red, HEX -> #ff0000 RGB -> rgb(255, 0, 0);	sets the color of an element's border.
border-radius	Absolute unit -> 4px, Relative unit-> 2em, 2%	used to add rounded borders to an element

```

<!DOCTYPE html>
<html lang="en">
<head>
  <style>
    p{
      border-width: 5px;
      border-style: solid;
      border-color: red;
    }
  </style>
</head>
<body>
  <p>Lorem ipsum dolor sit, amet consectetur adipisicing elit. Cum,
  laudantium.</p>
</body>
</html>

```

Shorthand of above properties

```
<style>
  p {
    border : 5px solid red;
  }
</style>
```

4. Margin

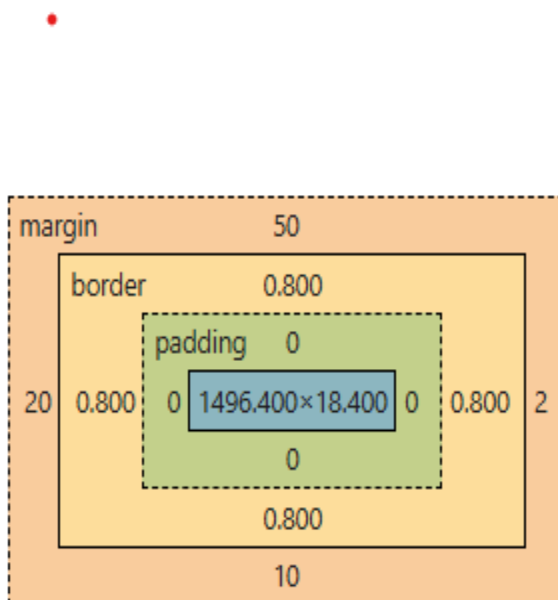
Margin is the outermost transparent layer that surrounds the border.

It creates space between this element and nearby elements, controlling how close or far they appear from one another.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <style>
    div{
      margin-top : 50px;
      margin-right: 2px;
      margin-bottom: 10px;
      margin-left: 20px;
      border : 1px solid black;
    }
  </style>
</head>
<body>
  <div>
    Lorem ipsum dolor sit amet consectetur adipisicing elit. Ipsa molestiae
    ea eum beatae odio quasi distinctio quo tempora deserunt. Provident qui minus
    eligendi in cupiditate iusto inventore praesentium maxime! Veritatis!
  </div>
</body>
</html>
```

Shorthand of margin properties :

```
<style>
  div{
    margin : 50px 2px 10px 20px;
    border : 1px solid black;
  }
</style>
```



Height :

1. Sets a **fixed height** for an element.
2. The element will **not grow or shrink** beyond this height — even if the content inside overflows.

Width :

1. Sets a **fixed width** of an element.
2. The element will **not become wider or narrower** than this — even if the content or parent changes.

Min-width :

1. Defines the **minimum width** of an element.
2. The element can **grow wider** if needed, but **never smaller** than this.

Min-height :

1. Defines the **minimum height** of an element.
2. The element can **grow taller** if content requires more space, but **never shrink below** this value.

Height and Width Example.

```
<style>
  div{
    height: 200px;
    width: 200px;
    border : 3px solid red;
  }
</style>
```

Lorem ipsum dolor sit amet
consectetur, adipisicing elit.
Illo cum, et voluptatum,
maiores quasi facilis ad
reprehenderit, labore illum
laudantium unde! Neque
provident commodi ipsum esse
officiis soluta mollitia
molestias distinctio, quod
laudantium rem explicabo
reiciendis doloribus sunt
facilis adipisci odio laborum
eveniet excepturi? Fugiat,
officia id voluptas debitis
deleniti, sit dicta dolore natus
distinctio minima sed pariatur
totam quod sunt assumenda
tempora? Repellendus
sapiente, maiores quia laborum
cupiditate dolorem sequi, iusto

Min-height and min-width example:

```
<style>
  div{
    min-height: 200px;
    min-width: 200px;
    border : 3px solid red;
  }
</style>
```

Lorem ipsum dolor sit amet consectetur, adipisicing elit. Illo cum, et voluptatum, maiores quasi facilis ad reprehenderit, labore illum laudantium unde! Neque provident commodi ipsum esse officii soluta mollitia molestias distinctio, quod laudantium rem explicabo reiciendis doloribus sunt facilis adipisci odio laborum eveniet excepturi? Fugiat, officia id voluptas debitis deleniti, sit dicta dolore natus distinctio minima sed pariatur totam quod sunt assumenda tempora? Repellendus sapiente, maiores quia laborum cupiditate dolorem sequi, iusto molestias illum facilis tempore ad similique quos est eligendi doloribus laboriosam! Magni, earum. Hic dolore ipsam voluptatum necessitatibus dolorum deserunt rem quibusdam iste saepe. Blanditiis illo optio praesentium eligendi officii autem porro repellat laboriosam placeat iste, sapiente necessitatibus itaque voluptas doloremque tempore consectetur eveniet tempora vel temporibus laborum ea. Maxime quasi et a nostrum perspiciatis quam, iste quod, laudantium asperiores quaerat dolorem modi eum? Accusamus, odio molestias expedita facere explicabo beatae doloribus magni quam ad, quis porro dolore officii possimus non illo culpa quidem architecto commodi. Animi deserunt asperiores exercitationem est expedita recusandae ea quis nihil? Nemo inventore nihil voluptas? Accusantium officii excepturi voluptatem iste, maiores, consectetur earum dicta aliquid accusamus ullam, ipsum soluta eos beatae laudantium assumenda commodi pariatur in suscipit voluptate unde vero sit? Illum nisi doloremque consequatur labore? Nam nesciunt dolore sequi eum aliquam, quae soluta voluptate cumque earum omnis? Explicabo sit quisquam, sunt eveniet sapiente eius ipsam non dolorem tempore nesciunt. Ut, sint est! Dolorum, preferendis quibusdam! Magnam odit ea dolorem corrupti harum odio? Aspernatur saepe voluptatem a explicabo, aliquam quis temporibus pariatur eos in accusantium quos dolorum non nesciunt dolor atque commodi molestias similique repellendus modi rem cumque, deserunt autem! Culpa adipisci facere facilis earum, deserunt numquam vitae amet harum? Commodi, cumque. Omnis harum voluptatum ab iste delectus veritatis placeat illum eum, aperiam id qui perspiciatis natus. Laudantium quam adipisci, enim amet earum sit labore, rerum illo tempora officia provident sint. Sit repudiandae expedita animi exercitationem molestiae minima velit! Provident rem odio aspernatur id eaque consequatur molestias vitae harum, accusantium aliquid. Magnam, nulla ipsam, voluptatibus eos beatae error, consequatur ab obcaecati iusto consequuntur rem sequi illo nemo placeat enim pariatur iste officia voluptatum deleniti corporis impedit natus animi a. Enim, similique quos! Beatae ad assumenda nesciunt illo et quidem rerum cupiditate voluptas neque expedita ut fugit quia impedit nobis nam facere, minima suscipit! Officia ut minima nihil. Provident hic officia, voluptatum eaque, nihil eligendi, commodi error quas esse soluta cum sunt voluptatibus placeat voluptates porro vero! Magnam mollitia incidunt, soluta quas aliquid, placeat nihil deleniti quod ipsam obcaecati nam illo illum dolor ducimus ipsa ipsum ullam provident ratione vitae? Asperiores dolor neque saepe nam reiciendis error iure laborum porro sunt ratione aliquam rerum debitis quisquam est beatae magnam ipsa, officia corrupti libero ducimus? Magnam possimus ullam omnis, ex pariatur ut, harum non blanditiis voluptate, nemo nostrum exercitationem dolore enim. Architecto quidem, corporis voluptatum labore natus quasi nisi accusamus sequi ipsum, sint ad aliquid error id quae dolor cumque officia expedita rem quo preferendis velit quos. Neque accusantium quod porro cum eligendi, eius molestias!

Gradient :

A CSS gradient is a type of image data type that allows you to display smooth transitions between two or more specified colors. Instead of using a static image file, the browser generates the gradient dynamically.

There are three main types of gradients in CSS:

1. Linear Gradient

Creates a straight-line color transition.

Default direction of gradient is top to bottom

Syntax :

background: linear-gradient(direction, color1, color2, ...);

```
<!DOCTYPE html>
<html>
<head>
<style>
#grad1 {
  height: 200px;
  background-image: linear-gradient( blue, yellow);
}
```



```
</style>
</head>
<body>

<h1>Linear Gradient - Top to Bottom</h1>
<p>This linear gradient starts blue at the top, transitioning to
yellow at the bottom:</p>

<div id="grad1"></div>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
<style>
#grad1 {
  height: 200px;
  background-image: linear-gradient(90deg, red, yellow, green
80%, blue);
}
</style>
</head>
<body>

<h1>linear gradient</h1>

<div id="grad1"></div>
```

```
</body>
</html>
```

Radial Gradient:

The **radial-gradient()** function in CSS makes a background that smoothly changes colors starting from the **center** and spreading **outward**.

It can be shaped like a **circle** or an **ellipse**.

It's treated like an **image** that you can use as a background in your webpage.

Note : if we don't provide anything in first parameter in radial-gradient then css will consider it as ellipse by default

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Circle vs Ellipse Gradient</title>
<style>
.container {
  display: flex;
  gap: 20px;
}
.box {
  width: 300px;
  height: 150px;
  border: 2px solid black;
```

```

}
.circle {
  background: radial-gradient(circle, yellow, red, aqua , green);
}

/* if we dont provide anything then css will consider ellipse by default */
.ellipse {
  background: radial-gradient( yellow, red,  aqua , green);
}
</style>
</head>
<body>
<div class="container">
  <div class="box circle">Circle</div>
  <div class="box ellipse">Ellipse</div>
</div>
</body>
</html>

```

Variable Spaced Color Stops :

```

<!DOCTYPE html>
<html>

<head>
  <style>
    #radial {
      height: 500px;
      width: 500px;
      background-image: radial-gradient(red 5%, yellow 15%, green
60%);
    }
  </style>
</head>

<body>

```

```
<h1>Radial Gradient - Differently Spaced Color Stops</h1>
<div id="radial"></div>
</body>

</html>
```

Conic Gradient :

The `conic-gradient()` CSS function creates an image consisting of a gradient where color transitions rotate around a center point, rather than radiating outwards (like a radial gradient) or along a straight line (like a linear gradient).

Key Characteristics :

- Rotation: Colors transition clockwise around the circumference of a circle, starting at the top (0 degrees).
- Center: The gradient uses the center of the element as the source point for the color stops by default.
- Color Stops: Unlike linear and radial gradients which use lengths, the color stops in a conic gradient are specified using angles (e.g., `deg`, `rad`, `turn`, `grad`).

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Conic Gradient Example</title>
<style>
.gradient-box {
  width: 250px;
  height: 180px;
  background: lightgray; /* fallback for unsupported browsers */
  background-image: conic-gradient(blue, cyan, lime, yellow);
  border: 2px solid #333;
}
```

```

    border-radius: 10px;
}
</style>
</head>
<body>

<h2>Conic Gradient with Four Colors</h2>

<div class="gradient-box"></div>

</body>
</html>

```

Conic gradient with angle Stops :

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Conic Gradient with Degrees</title>
<style>
.gradient-box {
    width: 250px;
    height: 250px;
    border-radius: 50%;
    border: 2px solid #333;
    background: conic-gradient(
        red 0deg 90deg,
        yellow 90deg 180deg,
        green 180deg 270deg,
        aqua 270deg 360deg
    );
}

```

```
</style>
</head>
<body>

<h2>Conic Gradient with Angle Stops</h2>

<div class="gradient-box"></div>

</body>
</html>
```

Shadows:

Shadow effects are used to add depth and highlight elements.

It has 2 properties:

text-shadow → for text

box-shadow → for boxes or elements

Text-shadow:

Syntax -

text-shadow : `offset-x | offset-y | blur-radius | color`

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  text-shadow: 3px 1px cyan;
}
</style>
</head>
<body>
```

```
<h1>Text-shadow</h1>

</body>
</html>
```

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  text-shadow: 0 0 3px rgb(6, 9, 218);
}
</style>
</head>
<body>

<h1>Text-shadow having blurr effect</h1>

</body>
</html>
```

Box Shadow :

box-shadow: h-offset v-offset blur spread color inset;

Component	Description	Required	Example
h-offset	Horizontal offset (position). Positive moves the shadow right; negative moves it left.	Yes	5px
v-offset	Vertical offset (position). Positive	Yes	10px

	moves the shadow down; negative moves it up.		
blur	The blur radius . Higher values create a softer, larger shadow. Default is 0.	No	8px
spread	The spread radius . Positive values expand the shadow's size; negative values shrink it. Default is 0.	No	2px
color	The color of the shadow. Best practice is to use <code>rgba()</code> or <code>hsla()</code> for transparency.	No	<code>rgba(0, 0, 0, 0.4)</code>
inset	Changes the shadow from a drop shadow (outer) to an inner shadow (inside the element's frame).	No	inset

```

<!DOCTYPE html>
<html>

<head>
  <style>
    .box {
      width: 200px;
      height: 100px;
      background: lightblue;
      box-shadow: 3px 3px 12px 4px rgba(0, 0, 0, 0.5);
      border-radius: 10px;
    }
  </style>

```



```
</head>

<body>

  <h2>Box Shadow</h2>
  <div class="box"></div>

</body>
</html>
```

Typography

Typography in CSS refers to using different properties to style and arrange text on a webpage. It helps improve **readability, aesthetics, and user experience** by defining the **font style, size, spacing, alignment, and decoration** of text.

Typography properties can be grouped into two main categories:

1. **Font-related properties**
2. **Text layout and spacing properties**

Font-Related Properties

These properties define the overall appearance of the font — such as its family, weight, and style.

Property Name	Vaues	Description
font-family	"Arial", "Times New Roman", "Poppins", sans-serif, serif, monospace	Defines the typeface (font) for text. You can specify multiple fonts with fallbacks.
font-weight	normal, bold, lighter, bolder, 100–900	Controls the thickness (boldness) of text.

font-size	12px, 1em, 1rem, 100%, small, large	Specifies the size of the text. Can be in absolute or relative units.
font-style	normal, italic, oblique	Defines whether text is shown normally, in italics, or slanted
font-variant	normal, small-caps	Displays text in small capital letters.
font-stretch	normal, condensed, expanded, ultra-condensed, extra-expanded	Adjusts the width of characters (not supported in all browsers).

Example:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Font Properties Example</title>
  <style>
    body {
      background-color: #f4f4f4;
      font-family: "Segoe UI", Arial, sans-serif;
      padding: 30px;
    }

    h1 {
      font-family: "Poppins", sans-serif;
      font-size: 40px;
      font-weight: 700;
      font-style: italic;
      font-variant: small-caps;
      color: #2c3e50;
    }

    h2 {
      font: bold 28px "Roboto", sans-serif;
```

```

        /* shorthand property */
        color: #34495e;
    }

    p {
        font-family: "Open Sans", Verdana, sans-serif;
        font-size: 18px;
        font-weight: 400;
        font-style: normal;
        font-variant: normal;
        color: #555;
    }

    .note {
        font: italic 500 16px/5.5 "Courier New", monospace;

        /* font-style: italic;
        font-weight: 500;
        font-size: 16px;
        line-height: 1.5;
        font-family: "Courier New", monospace; */

        color: #8e44ad;
    }
</style>
</head>

<body>

    <h1>Font Properties in CSS</h1>
    <h2>Understanding Font Styling</h2>
    <p>
        Font properties in CSS control how text appears, including its
        typeface, size, style, and weight.
        They allow designers to create visually appealing and readable content.
    </p>
    <p class="note">
        This paragraph uses the <b>font shorthand property</b> to combine
        style, weight, size, line height, and font
        family in one declaration.
    </p>

</body>

</html>

```

Text-Layout and Spacing Properties:

Property	Possible Values / Examples	Description
text-align	left, right, center, justify	Aligns text horizontally within its container.
text-decoration	none, underline, overline, line-through	Adds or removes text decorations like underlines or strikethroughs.
text-transform	none, capitalize, uppercase, lowercase	Controls the capitalization of text.
text-indent	20px, 2em	Adds indentation to the first line of a paragraph.
letter-spacing	normal, 2px, 0.1em	Adjusts space between individual letters.
word-spacing	normal, 5px, 1em	Controls space between words.
line-height	normal, 1.5, 20px	Sets the height of each line of text (vertical spacing).
white-space	normal, nowrap, pre, pre-wrap	Controls how white space inside text is handled.


```
.text-sample {
  font-family: "Georgia", serif;
  font-style: italic;
  font-variant: small-caps;
  font-weight: 600;
  font-size: 20px;
  line-height: 1.8;
  letter-spacing: 1px;
  word-spacing: 6px;
  text-align: justify;
  text-indent: 40px;
  text-transform: capitalize;
  text-decoration: underline;
  text-shadow: 2px 2px 4px gray;
  direction: ltr;
  white-space: normal;
  word-wrap: break-word;
  text-overflow: ellipsis;
  background-color: #fff;
  border-left: 6px solid #4a90e2;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0px 4px 10px rgba(0, 0, 0, 0.1);
}
</style>
</head>

<body>

  <h2>Typography and Text Layout Properties Example</h2>

  <p class="text-sample">
    css typography includes various font and text layout properties that
    control how text looks and behaves on a web
    page.
    by combining font style, size, weight, spacing, and alignment,
    designers can create visually appealing and
    readable content.
    properties like text-align, line-height, and letter-spacing improve
    layout, while font-family and font-weight
    enhance appearance.
  </p>

</body>
```

```
</html>
```

Web Safe Fonts:

Web-safe fonts are fonts that are pre-installed on most operating systems and devices, so they display consistently across different browsers and platforms — without needing to download external font files.

They ensure that your website's text looks the same on Windows, macOS, Android, iOS, etc.

Font Name	Font Family Type	Notes / Description
Arial	Sans-serif	Clean and widely used; safe on almost all systems.
Verdana	Sans-serif	Designed for screen readability; slightly wider letters.
Helvetica	Sans-serif	Popular on macOS; similar to Arial.
Tahoma	Sans-serif	Compact and easy to read on screens.
Trebuchet MS	Sans-serif	Friendly, modern appearance.
Times New Roman	Serif	Traditional newspaper-style font.
Georgia	Serif	Similar to Times but more readable on screens.
Garamond	Serif	Elegant and classic, great for print-style designs.
Courier New	Monospace	Typewriter-style; good for code or terminal text.
Lucida Console	Monospace	Clean, fixed-width font for code.

Palatino Linotype / Book Antiqua	Serif	Elegant serif fonts suitable for headings.
Impact	Sans-serif	Thick, bold font — good for titles.
Comic Sans MS	Sans-serif	Informal, friendly font (use sparingly).
Brush Script MT	Cursive	Handwritten-style script font.